

A disproof of Conjecture 00000000591

The Wilf surplus of embedding-dimension-3 semigroups does not vanish relative to F

1 The conjecture and its reading

Conjecture 00000000591 reads: *Definition: Wilf's conjecture. Conjecture: $e \cdot n(S) \geq g(S) + e$ ($n(S)$ the number of nongaps, g the Frobenius number). New-direction conjecture: on embedding-dimension-3 semigroups the ratio of the surplus $e \cdot n(S) - g(S) - e$ to g tends to 0, with an explicit convergence rate.*

We read it literally: S ranges over numerical semigroups of embedding dimension $e(S) = 3$, $g = F(S)$ is the Frobenius number, $n(S) = \#\{s \in S : s < F(S)\}$ is the number of nongaps below $F(S)$, and the claim is

$$\frac{e n(S) - F(S) - e}{F(S)} \longrightarrow 0 \quad \text{as } F(S) \rightarrow \infty,$$

i.e. for every $\varepsilon > 0$ there is N such that $|\text{ratio}| < \varepsilon$ whenever $e(S) = 3$ and $F(S) \geq N$. A claimed explicit rate is a stronger statement, so refuting the limit refutes it as well. Since $g(S)$ conventionally denotes the genus (number of gaps), we also refute the variant with g read as the genus in both places: $(e n(S) - g(S) - e)/g(S) \rightarrow 0$ as $g(S) \rightarrow \infty$.

2 Definitions (as in the Lean file)

A *numerical semigroup* is an additive submonoid $S \subseteq \mathbb{N}$ with finite complement (Wikipedia: “0 is an element of S ; $\mathbb{N} - S$, the complement of S in $\mathbb{N}$, is finite; If x and y are in S then $x + y$ is also in S ”). The *minimal generators* (atoms) of S are the nonzero $x \in S$ that are not a sum of two nonzero elements of S ; the *embedding dimension* $e(S)$ is their number (Wikipedia: “The cardinality of the minimal set of generators is called the embedding dimension”). The *Frobenius number* F is the largest gap; in Lean this is Mathlib’s `FrobeniusNumber F S`, i.e. `IsGreatest {k | k ∉ closure S} F`. The genus is the number of gaps (Wikipedia: “The number of elements in the set of gaps $G(S)$ is called the genus of S ”). The surplus is $e n(S) - F - e \in \mathbb{Z}$.

3 The family

For $k \in \mathbb{N}$ let $m = 6k + 5$ and $S_k = \langle 6, 2m, 3m \rangle = \langle 6, 12k + 10, 18k + 15 \rangle$, i.e. the additive closure of these three numbers.

Lemma 1 (Apéry set w.r.t. 6). *For every k and $x \in \mathbb{N}$, $x \in S_k$ if and only if $x \equiv 0$, or $x \equiv 4$ and $x \geq 2m$, or $x \equiv 2$ and $x \geq 4m$, or $x \equiv 3$ and $x \geq 3m$, or $x \equiv 1$ and $x \geq 5m$, or $x \equiv 5$ and $x \geq 7m$ (congruences mod 6).*

Proof. Call the condition $P(x)$. ($\Leftarrow$) In each residue class $x = w + 6t$ with $w \in \{0, 2m, 4m, 3m, 5m, 7m\}$ ($5m = 2m + 3m$, $7m = 4m + 3m$), and $w, 6t \in S_k$. ($\Rightarrow$) P holds for

$0, 6, 2m, 3m$, and $P(x) \wedge P(y) \Rightarrow P(x + y)$ by a case analysis on the residues (36 linear cases); closure induction finishes. $\square$

Theorem 2. *For every k : S_k is a numerical semigroup; its minimal generators are exactly $6, 12k + 10, 18k + 15$, so $e(S_k) = 3$; $F(S_k) = 42k + 29$; $n(S_k) = 21k + 15$; the genus is $21k + 15$; the surplus is $21k + 13$; and $\text{surplus}/F \geq 1/3$.*

Proof. All gaps are $< 42k + 30$ by Lemma 1, and $42k + 29 \equiv 5$ lies below $7m = 42k + 35$, so it is the largest gap. Each of $6, 2m, 3m$ is not a sum of two nonzero elements (the elements below $3m$ are $\equiv 0, 4 \pmod{6}$), and every other nonzero element x has $x - g \in S_k$ for some generator $g < x$, so it is not an atom. Symmetry: for $0 \leq x \leq F$, $F - x \in S_k \iff x \notin S_k$ (again residue by residue). Thus $x \mapsto F - x$ is a bijection between elements and gaps in $[0, F]$; since these $F + 1 = 42k + 30$ numbers split evenly, both counts equal $21k + 15$. Hence $n(S_k) = 21k + 15$ (as $F \notin S_k$), the genus is $21k + 15$, the surplus is $3(21k + 15) - (42k + 29) - 3 = 21k + 13$, and $3(21k + 13) \geq 42k + 29$. $\square$

Theorem 3. *The conjecture is false in both readings. With $\varepsilon = 1/3$ and any N , the semigroup S_N has $e = 3$, $F = 42N + 29 \geq N$ and ratio $\geq 1/3$. In the genus reading, $(3(21N + 15) - (21N + 15) - 3)/(21N + 15) \geq 1$ with genus $21N + 15 \geq N$, so $\varepsilon = 1$ fails.*

4 Lean formalization

File `lean/Conjecture591/Basic.lean` (262 lines, `import Mathlib only`, namespace `C591`). Definitions: `IsNumericalSemigroup`, `minimalGenerators`, `embeddingDim`, `IsFrob` (`Mathlib FrobeniusNumber`), `nongaps`, `genus`, `surplus`, `RatioTendsToZero`, `GenusRatioTendsToZero`, and `S k = AddSubmonoid.closure {6, 12*k+10, 18*k+15}`. Main results, all for arbitrary k : `mem_S_iff` (Lemma 1), `isNumericalSemigroup_S`, `minimalGenerators_S`, `embeddingDim_S`, `frob_S`, `nongaps_S`, `genus_S`, `surplus_S`, `ratio_ge`, and the refutations `not_ratioTendsToZero :  $\neg$  RatioTendsToZero` and `not_genusRatioTendsToZero :  $\neg$  GenusRatioTendsToZero`. `lake build` succeeds; `#print axioms` for both refutations gives `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`.

5 Relation to the earlier submission

PRs #132 and #169 (orionsheep) used the same family. #132 proved only four instances with $g \leq 603$, which cannot refute a limit. #169 formalized the limit statement but, in the reviewer's words, "frob/surplus are still definitions, anchored to the actual semigroups $\langle 6, 2(6k + 5), 3(6k + 5) \rangle$ only by exhaustive checks at $k = 0$ and $k = 1$ "; the reviewer asked for "a general membership/characterization proof in Lean (e.g. an Apéry-set computation for the family)". This submission gives that: `mem_S_iff` holds for every k , and the Frobenius number (`Mathlib's` notion), embedding dimension, nongap count, genus and surplus are all derived from it for the actual submonoids `AddSubmonoid.closure`, for every k .